Meaning in Motion: *Trade* as Commerce, Craft, and Imperial Geography in a Historical Corpus

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Abstract

In this project, I analyze how the concept word *trade* varies across a structured historical corpus using named entity recognition, contextual embeddings, and supervised classification. Rather than treating *trade* as a stable lexical item, I approach it as a concept whose meaning depends on surrounding context, textual genre, and historical framing. After comparing candidate concept words, I selected *trade* because it was both frequent and semantically mobile. In the final analytic dataset, the term appeared 7,763 times across 341 documents. I then extracted local context windows around each occurrence, used spaCy to identify associated entities, embedded the contexts with a BERT-based model, and trained a logistic regression classifier to distinguish two recurring senses: *commercial exchange* and *occupation or craft*. The results show that *trade* is not distributed evenly across meanings. Commercial usage is overwhelmingly dominant and is strongly associated with geography, maritime movement, colonial exchange, and state scale. By contrast, the occupational or craft sense is present but far less central, and it emerges more clearly through supervised classification than through unsupervised clustering. I argue that the corpus organizes *trade* hierarchically: first as a broad semantic field centered on commerce, and then internally by subdomains such as shipping, foreign exchange, and imperial geography. This makes *trade* a useful case for showing that meaning variation in a historical corpus is structured, uneven, and method-dependent.

1 Introduction

The central question of this project is how meaning is constructed and varied across texts. Following the assignment prompt, I focus not on whole documents but on a single concept word and ask how its meaning changes across local contexts, how those changes relate to people and places, and which methods are best suited to making that variation visible. The structured version of the project requires a concept-centered analysis using named entity recognition, contextual embeddings, supervised classification, and close reading, with the final report integrating those methods into a single interpretive claim.

I selected *trade* as my concept word because it offered the strongest combination of frequency, contextual diversity, and interpretive payoff. In ordinary language, *trade* can refer to exchange and commerce, but it can also refer to a profession, craft, or calling. That duality made it a promising candidate for a project centered on semantic variation. At the same time, because the

corpus is historical and includes texts shaped by early modern spelling, imperial history, religion, and political economy, I expected the commercial meaning of *trade* to develop in ways that were more specific than a simple modern definition would suggest. My aim was therefore not just to show that *trade* has multiple meanings, but to determine how those meanings are distributed and structured in the corpus.

The argument that emerges from my analysis is that *trade* does not circulate in this corpus as a flat or evenly balanced concept. Instead, it is dominated by the semantic field of commercial exchange, especially in contexts involving geography, shipping, colonial movement, and state-scale economic thinking. A secondary meaning of *trade* as occupation or craft remains present and recoverable, but it is less central and less visible to unsupervised methods. Unsupervised contextual clustering primarily subdivides the dominant commercial meaning into several related subtypes, while supervised classification more clearly separates commercial uses from occupational ones. Taken together, these methods suggest that meaning variation is real but not symmetrical: some senses organize the semantic space of the corpus, while others survive as smaller but still interpretable branches.

2 Literature Review

Recent scholarship in computational text analysis is useful for my project not simply because it offers more powerful models, but because it clarifies what is difficult about studying meaning in historical corpora in the first place. Historical texts are noisy, heterogeneous, and often resistant to off-the-shelf NLP tools. As Ehrmann et al. argue in their survey of named entity recognition for historical documents, tasks that may appear routine on modern corpora become more unstable when spelling varies, OCR introduces noise, punctuation is inconsistent, and reference practices shift across time. That point is central to my own project. I use named entity recognition to study how *trade* is associated with places, institutions, and political actors, but the historical character of the corpus means that I have to interpret NER output cautiously. In other words, the literature suggests that historical NLP is valuable not because it eliminates interpretive judgment, but because it provides structured signals that still require historical reading.

A second relevant strand of scholarship comes from work on conceptual history and semantic change. Wevers and Koolen describe "digital begriffsgeschichte" as an effort to scale the study of concepts by using computational representations to trace how words move across semantic fields in historical corpora. Their argument is especially useful for my project because it frames computational methods not as replacements for close reading but as tools for expanding it. My task is not to define *trade* once and then count examples. It is to ask how different historical uses of *trade* cluster, separate, and overlap across a large collection of texts. In that sense, my project fits a broader methodological shift: using computational models to move from isolated examples toward corpus-level patterns of conceptual variation, while still treating those patterns as historically interpretable rather than self-explanatory.

That question of variation also explains why contextual embeddings matter here. Devlin et al. introduced BERT as a bidirectional transformer model that produces context-sensitive token rep-

resentations rather than assigning a single fixed vector to every word type. For a concept like *trade*, that distinction is crucial. Static representations can tell me that the word is important in the corpus, but they are less well suited to distinguishing between *trade* as commercial exchange and *trade* as occupation or craft. Work by Giulianelli, Del Tredici, and Fernández pushes this logic further by showing that contextualized word representations can be used to analyze lexical semantic change through clusters of usage types. That usage-based view is directly relevant to my project. Although I am not measuring change across two historical periods in a formal diachronic design, I am using the same basic insight: the meaning of a concept is better modeled through distributions of local usages than through a single abstract definition.

Finally, the literature on weak supervision helps justify the classification stage of my project. Ratner et al. argue that heuristic labeling can be a practical and defensible alternative when large amounts of manual annotation are infeasible. That framework fits this assignment well. The corpus is too large for exhaustive hand-labeling, but it is still possible to construct a weakly supervised training set that operationalizes a meaningful distinction between *commercial exchange* and *occupation or craft*. Taken together, these sources define the methodological logic of my paper. Ehrmann et al. explain why historical corpora make even basic extraction tasks uncertain; Wevers and Koolen show why computational approaches are useful for conceptual history; Devlin et al. and Giulianelli et al. provide a framework for modeling meaning at the level of contextual usage; and Ratner et al. show how imperfect but systematic supervision can support classification. This scholarship therefore does more than provide background. It establishes why a multi-method approach is necessary for studying how *trade* is structured across a noisy historical corpus.

3 Concept Selection, Corpus Preparation, and Unit of Analysis

I began by comparing several candidate concept words that appeared likely to support the full assignment pipeline. Among the strongest options were *credit*, *wealth*, *authority*, *liberty*, and *trade*. I did not want to choose a word simply because it was common. Instead, I looked for a concept that met four conditions: it had to appear frequently across the corpus, occur in many documents rather than a small handful, show evidence of contextual variation, and plausibly support at least two defensible label categories for supervised classification. *Trade* emerged as the strongest candidate because it combined very high frequency with an intuitively meaningful split between commercial exchange and occupation or craft.

In an initial selection pass, *trade* appeared 7,812 times across 342 documents. After final occurrence extraction and filtering for the analytic dataset, I retained 7,763 occurrences across 341 documents. That scale was large enough to support contextual embedding and classification, but still manageable for an all-occurrences pipeline. The distribution was also broad rather than narrowly concentrated, though several texts contributed especially large numbers of instances. The most prominent included *A10821.txt*, *Wealth_Nations.txt*, *A68617.txt*, *A71306.txt*, and *A71305.txt*. This pattern already suggested that *trade* was not confined to one isolated genre or authorial context.

For the unit of analysis, I used a local context window centered on each occurrence of *trade*. Rather

than analyze entire documents, I extracted the sentence containing the word along with immediately adjacent context when available. This design balanced interpretability and specificity. It preserved enough textual material for BERT-based contextual representation and logistic regression, while remaining close enough to the target word to support plausible sense distinctions. That choice also aligned with the core logic of the assignment: the project is not about broad document topic modeling but about how a concept behaves locally across many different textual settings.

The initial patterns confirmed that *trade* was semantically promising. Some contexts clearly framed it through merchants, commodities, countries, ports, coasts, and companies. Others used it in a more personal and occupational register, closer to a craft, profession, or social role. Even before formal modeling, then, the corpus suggested both polysemy and imbalance. The commercial meaning seemed much more common, while the occupational meaning appeared smaller but still legible.

4 Named Entity Recognition and the Geography of Trade

To understand how the corpus situates *trade* in relation to people, institutions, and places, I ran spaCy named entity recognition over the occurrence contexts. Because *trade* seemed likely to be tied to spatial and geopolitical language, I treated place-based entities as my main interpretive focus while still retaining counts for other entity types. The overall entity distribution was headed by `ORG`, `GPE`, and `PERSON`, each of which appeared at high frequency across the extracted contexts. That distribution is already suggestive. It implies that discussions of *trade* are rarely abstract in a purely conceptual sense; they tend to be attached to named institutions, named polities, and identifiable actors.

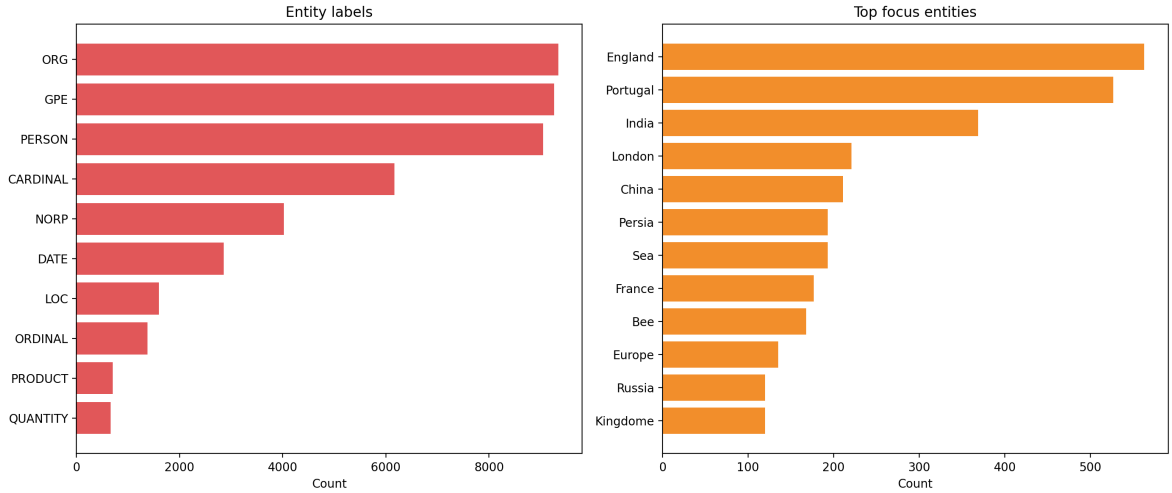

Figure 1: Named entity patterns in local contexts around *trade*. The left panel shows the overall distribution of extracted entity labels, while the right panel shows the most frequent focus entities.

The focus-entity table makes the spatial logic of *trade* even clearer. The most frequent entities include England (563), Portugal (527), India (369), London (221), China (211), Persia (193), France

(177), and Sea (193). Even allowing for occasional tagging errors, this is a remarkably coherent group. The dominant commercial uses of *trade* are not merely about exchange in the abstract; they are embedded in a world of routes, states, ports, imperial regions, and maritime circulation. England and Portugal point toward state and imperial competition. India, China, and Persia place trade in a transregional frame. London anchors part of the discourse in an urban and metropolitan center. Sea, while not a state, reinforces the importance of transport, route-making, and naval or commercial movement.

These NER results help sharpen the difference between the corpus's main semantic tendencies. When *trade* appears in commercial contexts, it is frequently tied to named places and collective political actors. The concept becomes geopolitical. It refers not only to exchange, but to exchange as something routed across territories, governed through imperial or national structures, and imagined at the scale of kingdoms, colonies, and long-distance commerce. By contrast, the occupational sense of *trade* tends to be less tightly organized around named places. It is more likely to refer to a kind of activity or social role than to a globally connected economic geography.

At the same time, the limitations of off-the-shelf NER matter here and should be stated explicitly. This corpus contains OCR noise, inconsistent historical spelling, and unstable punctuation. Those factors reduce the reliability of sentence segmentation and entity recognition. Some entities are almost certainly missed, and some tags are likely imperfect. For that reason, I do not treat the NER output as complete ground truth. Instead, I use it as a strong frequency signal and an interpretive aid. Even under noisy conditions, the entity results are too coherent to dismiss: they consistently show that commercial *trade* is strongly associated with geopolitical and maritime space.

5 Contextual Meaning with BERT

To model semantic variation more directly, I generated contextual embeddings for each occurrence window using a BERT-based representation and then reduced those embeddings to a two-dimensional space for visualization and clustering. I clustered the resulting points into four groups and used the nearest contexts and top terms within each cluster to interpret their thematic structure.

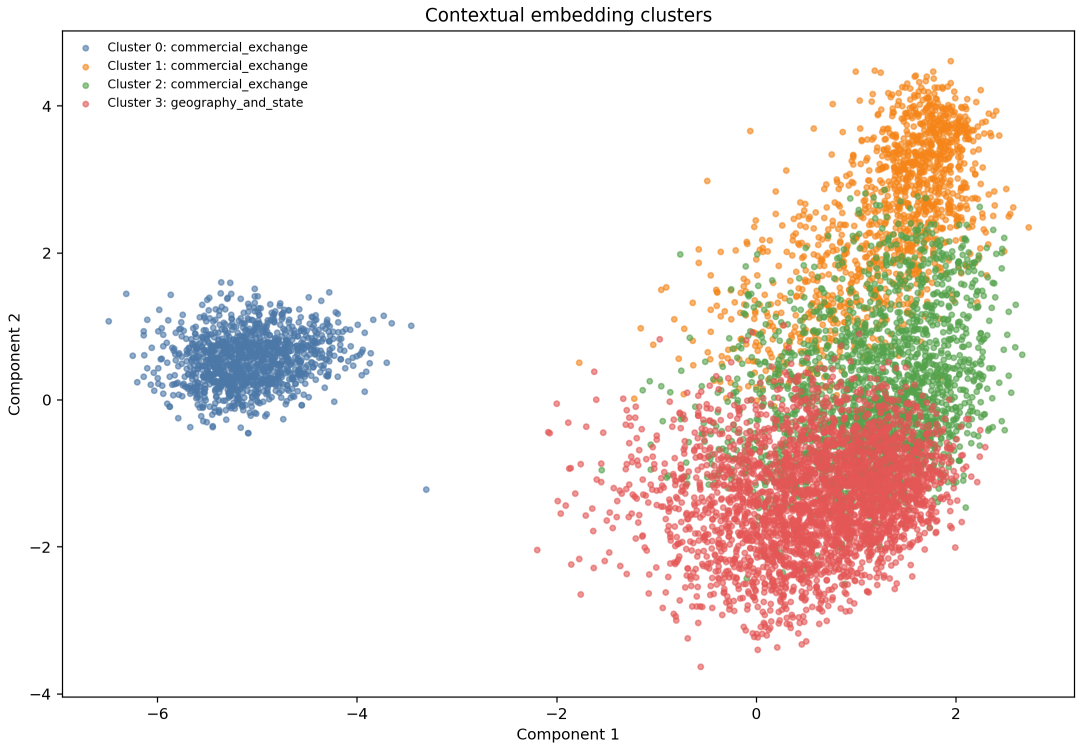

Figure 2: Contextual embedding clusters for occurrences of *trade*. One cluster is visibly separated from the others, while the remaining groups subdivide the dominant commercial field.

The first striking result is that the embedding space is not split into two equally sized semantic camps. Instead, three clusters remain fairly close to one another and are all most naturally interpreted as variants of commercial exchange, while one more isolated cluster appears to correspond to a different semantic pattern. In my cluster summary, Cluster 0 is characterized by terms such as *ship*, *country*, *Portugal*, *merchant*, *town*, and *company*. Cluster 1 emphasizes *foreign*, *citie*, *consumption*, and *merchandise*. Cluster 2 includes *colony*, *english*, *foreign*, *east*, *merchants*, and *commodities*. Cluster 3 includes *merchants*, *commodities*, *sea*, *ship*, *english*, *country*, and *east*, and I labeled it *geography and state*.

The clustering result matters because it complicates a simplistic reading of polysemy. If the corpus were divided evenly between commercial and occupational meanings, I would expect the unsupervised representation to separate those meanings in a more balanced way. Instead, the embedding space suggests that commercial *trade* is the dominant semantic center of gravity. What the unsupervised method mostly recovers is not a commerce-versus-craft distinction, but internal differentiation *within* commercial discourse. Some contexts emphasize maritime movement and shipping, others colonial or foreign exchange, and others a broader state or geopolitical frame. In other words, the BERT representation treats commercial *trade* not as a single stable meaning but as a layered field with recognizable internal structure.

That result is valuable precisely because it does not reproduce the final classifier labels in a mechanically perfect way. It shows that unsupervised contextual semantics and supervised sense

distinctions do not answer exactly the same question. The clustering procedure asks which contexts look most similar to one another in the embedding space. Under that criterion, the corpus's dominant semantic variation lies mostly inside the commercial register. The occupational or craft sense remains present, but it is less central to the geometry of the full dataset.

6 Supervised Classification of Trade Meanings

Because the assignment requires a classifier that distinguishes at least two meanings or uses of the concept, I next trained a logistic regression model to separate *commercial exchange* from *occupation or craft*. I began with weak labeling, generating an initial pool of candidate examples for each class. The raw weak-label pool was highly imbalanced: 3,921 examples were assigned to commercial exchange and 499 to occupation or craft. Rather than let the model simply learn the class imbalance of the corpus, I built a balanced training set with 250 examples per class. This decision makes the reported metrics more meaningful as a test of discrimination rather than prevalence.

On the held-out evaluation set, the classifier reached an accuracy of 0.736 and a macro F1 score of 0.736. Precision and recall were also reasonably balanced across the two classes. For *commercial exchange*, precision was 0.750 and recall was 0.714. For *occupation or craft*, precision was 0.723 and recall was 0.758. The confusion matrix shows 45 commercial examples correctly classified, 47 occupational examples correctly classified, and a total of 33 cross-class errors. Given the difficulty of the task, the historical noisiness of the corpus, and the fact that the labels are weakly inferred rather than fully hand-annotated, this is a credible and useful result.

The most informative aspect of the classifier is not the metric alone but the feature profile. Among the strongest features for *commercial exchange* are *east*, *great*, and *coast*. Those terms align closely with the NER and BERT results. They tie commercial *trade* to large-scale geography, maritime language, and long-distance movement. By contrast, terms such as *like*, *people*, and *haue* help push contexts toward the *occupation or craft* category. These are less obviously domain-specific, which itself is meaningful. The occupational sense often appears in more local, descriptive, or socially embedded formulations, rather than in the heavily spatialized discourse that dominates commercial usage.

The model also clarifies the asymmetry that the unsupervised analysis suggested. The occupational meaning is real enough to classify, but it does not dominate the semantic landscape of the corpus. It has to be surfaced more deliberately. That is, a supervised model trained to find this distinction can recover it with moderate success, but the full contextual embedding space is still mostly organized by variation inside the much larger commercial register.

There are, however, important limitations to the classifier. Weak labels introduce some circularity because the categories are initially inferred from lexical cues. I mitigated that risk by masking direct rule keywords before fitting, but the problem cannot be removed entirely. In addition, logistic regression produces a simplified linear boundary between senses that may blur genuinely ambiguous examples. Some contexts likely sit between the two categories or use *trade* in a way

that is only fully interpretable at a larger discourse scale. For those reasons, I treat the classifier as an interpretive instrument rather than a final arbiter of meaning.

7 Synthesis: What Meanings of *Trade* Exist, and How Are They Structured?

The most important task in this project is to move beyond individual outputs and make a single interpretive claim that integrates NER, contextual embeddings, classification, and close reading. My central claim is that *trade* in this corpus is organized hierarchically rather than evenly. The concept contains at least two recurring meanings, but those meanings do not occupy the corpus with equal force. Commercial exchange is the dominant semantic frame. Within that frame, the corpus further differentiates trade by geography, shipping, colonial position, and state-scale economic relation. The occupational or craft sense remains present, but it is secondary: visible in local contexts and recoverable through supervised classification, yet less central to the global semantic structure of the dataset.

Each method contributes something distinct to that argument. NER shows that discussions of *trade* are strongly attached to named geopolitical spaces such as England, Portugal, India, China, France, and London, along with the maritime marker *Sea*. This immediately situates the dominant meaning of *trade* in a world of routes, territories, and imperial circulation. BERT-based contextual clustering then shows that commercial *trade* is not semantically uniform. It has substructures: shipping and movement, foreign or colonial exchange, and broader state-geographic organization. Finally, logistic regression shows that another meaning of *trade*—occupation or craft—can be distinguished from the commercial field, even though it is numerically smaller and less geometrically dominant in the embedding space.

What close reading adds is a way of understanding why these methods diverge. In contexts where *trade* appears alongside merchants, commodities, coasts, companies, foreign regions, or named polities, the concept expands outward. It points to systems of circulation and exchange that are collective, geopolitical, and often imperial in scale. In contexts where *trade* refers to a person's trade, a practical calling, or a social role, the concept contracts inward. It becomes attached to labor, habit, or ordinary occupation rather than to transregional movement. These are not merely two dictionary definitions floating side by side. They are two differently structured ways of organizing social life. One imagines trade as a networked economic system; the other imagines it as a form of situated human work.

This is why the difference between the BERT result and the classifier result is not a weakness but an insight. The unsupervised model reveals where the corpus is semantically dense. That density lies mostly within commerce. The supervised model, by contrast, reveals a distinction that matters interpretively even if it is not the principal organizing axis of the full embedding space. Together, the two results show that meaning variation cannot be reduced to a single computational procedure. Different methods detect different layers of structure. In this corpus, *trade* is polysemous,

but its polysemy is unevenly distributed and historically weighted toward commerce, empire, and geography.

8 Limitations

Several limitations shape the interpretation of this project. First, the corpus itself is noisy. OCR artifacts, variable spelling, and unstable punctuation affect sentence segmentation, local context extraction, and named entity recognition. Second, the classifier depends on weak labels, which means the two target meanings are not based on a fully manual gold-standard annotation set. Third, the cluster labels are interpretive rather than intrinsic. BERT produces a geometric representation of contextual similarity, but the meaning assigned to a cluster still depends on human reading of representative examples and top terms.

These limitations do not invalidate the project, but they do set appropriate boundaries on the claims I can make. I do not claim that the NER table is exhaustive, that the clusters are final semantic truths, or that the classifier captures every nuance of *trade*. Instead, I claim that across multiple methods, a stable interpretive pattern emerges: the corpus overwhelmingly organizes *trade* around commercial exchange, and within that dominant meaning it develops important substructures tied to geography and state scale, while a smaller occupational sense remains visible at the margins.

9 Conclusion

This project began with a simple question: how is meaning constructed and varied across texts? By centering the analysis on *trade*, I found that meaning variation in this corpus is real but not flat. *Trade* does not function as a single stable concept, nor does it divide neatly into two equally weighted senses. Instead, it is structured asymmetrically. The dominant semantic field is commercial and strongly linked to space, maritime movement, foreign exchange, and political geography. A secondary sense of *trade* as occupation or craft remains interpretable, but it is less central and becomes clearest when I apply supervised classification.

Methodologically, the project also shows why using multiple NLP approaches is valuable. Named entity recognition reveals the geopolitical environment of the concept. Contextual embeddings reveal internal variation within the dominant meaning. Supervised classification recovers a less frequent but still important contrast. Close reading then connects those signals back to historically legible uses of the term. The result is not just a description of a word, but an account of how a concept is distributed across a corpus: unevenly, hierarchically, and in ways that reflect broader historical forms of economic and political imagination.

References

[1] Devlin, Jacob, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 4171–4186.

[2] Ehrmann, Maud, Ahmed Hamdi, Elvys Linhares Pontes, Matteo Romanello, and Antoine Doucet. 2023. "Named Entity Recognition and Classification in Historical Documents: A Survey." *ACM Computing Surveys* 56, no. 2.

[3] Giulianelli, Mario, Marco Del Tredici, and Raquel Fernández. 2020. "Analysing Lexical Semantic Change with Contextualised Word Representations." *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 3960–3973.

[4] Ratner, Alexander, Stephen H. Bach, Henry Ehrenberg, Jason Fries, Sen Wu, and Christopher Ré. 2020. "Snorkel: Rapid Training Data Creation with Weak Supervision." *The VLDB Journal* 29, no. 2: 709–730.

[5] Wevers, Melvin, and Marijn Koolen. 2020. "Digital begriffsgeschichte: Tracing semantic change using word embeddings." *Historical Methods: A Journal of Quantitative and Interdisciplinary History* 53, no. 4: 226–243.